

- Please note that these are sample questions for practicing on the concepts and not questions from the exam

Q

- a. (2 pnts) Handling the imbalanced data can be done in different ways, list the different techniques that can be used for this purpose. Which one of them is the least feasible technique and why?
- b. (2 pnts) Given:
 - the number of positive/negative examples in a dataset is 600/300
 - 200 examples from the unprivileged group with 120 positive examples
 - 480 positive examples from the privileged group.
 - We have only privileged/ unprivileged but nothing in between.

Compute the disparate impact ratio and comment on whether this dataset is fair or not and why? (Note: the formula of DI-ratio metric will be given in the exam if you have a question that requires it or the other fairness metrics).

Q

- c. (2 pnts) How does SMOTE work to generate synthetic examples? Use SMOTE to generate one more example between the points $x_1 = (a_1, b_1)$ and $x_2 = (a_2, b_2)$
- d. (2 pnts) Explain how generative adversarial networks work and list two of the failure cases.

A

- a. (2 pnts) Handling the imbalanced data can be done in different ways, list the different techniques that can be used for this purpose. Which one of them is the least feasible technique and why?
- **Answer:**
 - Imbalanced data can be handled with:
 - Collecting more sample for the under-represented class
 - Over/under sample the data
 - Generate synthetic data
 - Use cost sensitive classifier
 - Use ensemble of classifiers
 - Which one of them is the least feasible technique and why?
 - The least feasible technique is collecting more samples as this requires a long time and there is no control on how to get more examples from the under-represented class.

A

- b. (2 pnts) Given:
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 - We have only privileged/ unprivileged but nothing in between.

Compute the disparate impact ratio and comment on whether this dataset is fair or not and why? (Note: the formula of DI-ratio metric will be given in the exam if you have a question that requires it or the other fairness metrics).

Answer: $P(Y(x) = 1 \mid S = G) = \frac{120}{200} = 0.6$ and

$$P(Y(x) = 1 \mid S = G') = \frac{480}{700} = 0.686$$

$$\text{Hence } DI_{ratio} = \frac{0.6}{0.686} = 0.875$$

This dataset is said to be fair with respect to the DI-ratio as the requirement is that $0.8 \leq DI_{ratio} \leq 1.25$.

A

- c. (2 pnts) How does SMOTE work to generate synthetic examples? Use SMOTE to generate one more example between the points $x_1 = (a_1, b_1)$ and $x_2 = (a_2, b_2)$
- Solution:
- SMOTE work by picking a sample from the minority class randomly (say x), finds its k-nearest neighbors and pick one them (say y). Then it picks a random number in the interval $\delta \in [0, 1]$ and multiply the difference between x and y by δ and add the difference to one the sample (x or y) to generate a new example x_g . For example, $x_g = \delta(x - y) + x$
- To generate a new example from $x_1 = (a_1, b_1)$ and $x_2 = (a_2, b_2)$, we pick a number between 0 and 1 (let us say 0.4) and compute
 - $x_g = (0.4 * (a_1 - a_2) + a_1, 0.4 * (b_1 - b_2) + b_1)$

A

- d. (2 pnts) Explain how generative adversarial networks work and list two of the failure cases.
- Solution:
- Generative adversarial networks consists of two neural networks (generator and discriminator). The two networks compete in a zero-sum game approach, where the generator generates new samples that are similar to the original data and the discriminator classifies the samples as real or fake.
- Failure cases:
 - The discriminator become very strong and correctly identifies all the generated examples
 - The generator learns only a few specific weaknesses of the discriminator

Q2

- 2.1. (2 pnts) Explain how z-score normalization works and use it to normalize the data $D = \{1, 5, 9, 13, 17, 21, 25\}$.
- $\text{Mean}(D) = 13$
- $\text{Std}(D) = 8$
- $\text{Normalized}(D) = \{-1.5, -1, -0.5, 0, 0.5, 1, 1.5\}$



Q2

- 2.2. (2 pnts) If you have the following data: $D = \{2, 5, 77, 6, 15, 30, 8, 17, 45, 25, 60, 28\}$, what will be the contents of the 3 bins that you will use to smooth the data by bin boundaries?
- Answer:
 - Since it didn't mention what technique should we use, we can select any technique
- 2.3. (2 pnts) Describe the parametric statistical approach for outlier detection and use the assumption that the data is normally distributed to detect the outliers in the following dataset:
 $D = \{2, 5, 3, 6, 4, 16, 3, 4, 2, 3, 7, 5\}$.



Q2

- 2.2. (2 pnts) If you have the following data: $D = \{2, 5, 77, 6, 15, 30, 8, 17, 45, 25, 60, 28\}$, what will be the contents of the 3 bins that you will use to smooth the data by bin boundaries?
- **Answer:**
 - Since it wasn't mentioned, which technique we should use, we can select any technique
 - I will use both:
 - Equi-depth binning: We sort the data to get $DS = \{2, 5, 6, 8, 15, 17, 25, 28, 30, 45, 60, 77\}$ then
 - $B1 = \{2, 5, 6, 8\}$, $B2 = \{15, 17, 25, 28\}$, $B3 = \{30, 45, 60, 77\}$
 - $B1s = \{2, 2, 8, 8\}$, $B2s = \{15, 15, 28, 28\}$, $B3s = \{30, 30, 77, 77\}$
 - Equi-Width binning: We will use the DS also: $\text{bin width} = (\max(D) - \min(D) + 1) / \#bins = 76 / 3 = 25.33$
 - $B1 = \{2, 5, 6, 8, 15, 17, 25\}$ elements in $[2, 27.33)$ then $B1s = \{2, 2, 2, 2, 25, 25, 25\}$
 - $B2 = \{28, 30, 45\}$ elements between $[27.33, 52.66)$ then $B2s = \{28, 28, 45\}$
 - $B3 = \{60, 77\}$ elements between $[52.66, 78)$ then $B2s = \{60, 77\}$



Q2

- 2.3. (2 pnts) Describe the parametric statistical approach for outlier detection and use the assumption that the data is normally distributed to detect the outliers in the following dataset:

$$D = \{2, 5, 3, 6, 4, 16, 3, 4, 2, 3, 7, 5\}.$$

Answer:

With the normality assumption (data is extracted from normal distribution), we consider a value as outlier if the distance from the value to the mean of the data is greater than 3 times the standard deviation of the data. That is

Let $m = \text{mean}(D)$ and $s = \text{std}(D)$, $\forall x \in D$, if $|x - m| > 3 * s$ then outlier.

For the given data: $m = 5$, $s = 3.63$

Outliers = {16}

